



SOLAR PHOTOVOLATIC TECHNOLOGY – LOAD (W) and GENERATION (W) PREDICTION PAKISTAN

Submitted to:

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1. Introduction

1.1 Project Overview

Solar power has emerged as a crucial component of the global energy landscape in pursuit of climate change and due to the urgent need for sustainable energy solutions. Our project mainly focuses on developing efficient machine learning models for the prediction of photovoltaic (PV) generation and load consumption. We used historical data to make energy management within power grids more efficient. We selected two models to develop: LightGBM and LSTM. The reason to choose them is their effective use in time-series forecasting. It is important to make accurate predictions for optimizing the distribution of renewable energy sources as it helps reduce operational costs and promotes sustainable energy systems.

We pursued this project because it has the potential to improve operational efficiency and has broader implications for achieving sustainable energy, which is a crucial need in Pakistan. By producing precise estimates, our effort is directed towards reducing waste, maximizing resource allocation, and ultimately helping reduce greenhouse gas emissions. This project makes all of this possible while working in line with international efforts to improve power system resilience and switch to greener energy sources.

1.2 Purpose and Scope

Our project's primary goal is to create models that predict energy generation and demands in order to provide scalable real-time grid management. To cater to this goal to the best of our abilities, we went through a number of essential and demanding steps (feature engineering, data preprocessing, and model construction) in order to ensure the accuracy and efficiency of the predictions.

The project's scope consists of:

- 1) **Data collection:** First we compiled the relevant weather data and historical data on PV generation and load usage.
- 2) **Data preprocessing:** Then we moved to this step to address the missing values, outliers, and duplicates as well as clean and prepare the data for the model's training.
- 3) **Feature engineering:** We then used this process to develop features that enhance model performance and highlight the subtleties of the data.
- 4) **Model Development:** In our model development stage we implemented LightGBM and GRU models for optimized and accurate predictions.
- 5) **Performance Evaluation:** We reviewed both of these models, using KPIs that we narrowed down, to offer recommendations for further development.

By implementing this thorough and perceptive method, we strive to produce effective energy management decision-making, which will enhance sustainability and efficiency in utilizing renewable energy sources.

2. Methodology

2.1 Data Collection

The historical data sets that we used were from two primary sources: PV Generation and Load Consumption Records. In addition to these, we sourced meteorological data from Metrostate to draw connections between power generation and variables such as temperature, windspeed, and atmospheric pressure. It was essential to use this weather data to predict the variability of solar energy generation, because solar output is highly dependent on atmospheric conditions.

After collecting the datasets we organized them with timestamps and system identifiers to ensure synchronization across records. This was an essential step because the models that we used are dependent on time-dependent patterns in the data. As the data spanned across multiple years, we had a rich source of information to train the machine learning models.

2.1.1 Dataset Overview

The dataset covers one year of data at 10-minute intervals from multiple PV systems.

The goal is to predict future PV generation and load consumption based on historical data. Features
Timestamp: Date and time of the data sample (e.g., YYYY-MM-DD HH:MM:SS). System IDs: Unique identifiers for each PV system.

Location: City where the PV system is installed.

Connection Type: Type of system (Commercial or Residential).

Panels Capacity: Capacity of solar panels (kW).

Load Capacity: Capacity of the supported load (kW).

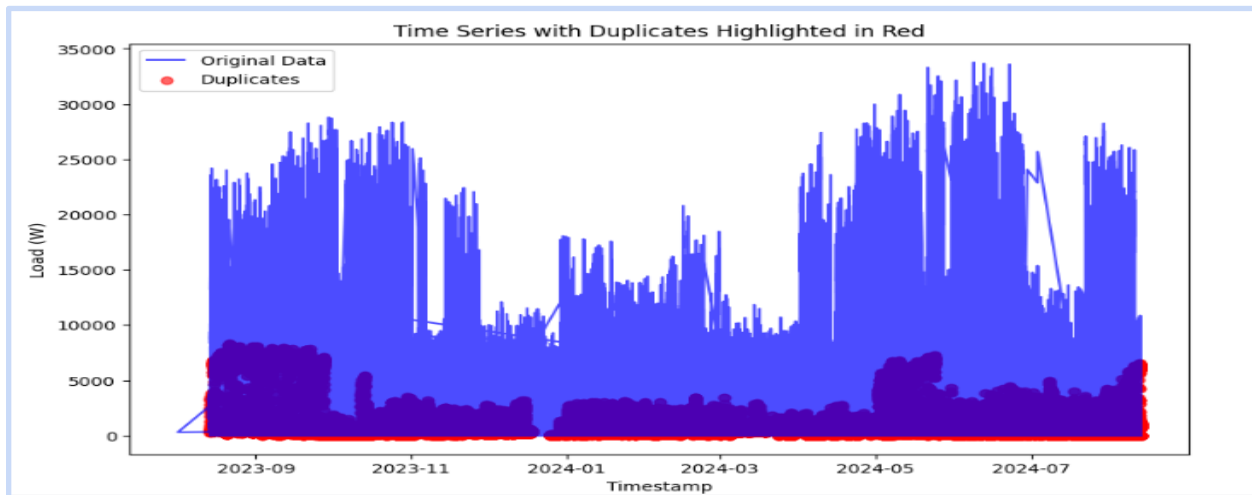
PV Generation (W): Power generated by the PV system (Watts).

Load Consumption (W): Power consumed by the load (Watts).

2.2 Exploratory data Analysis

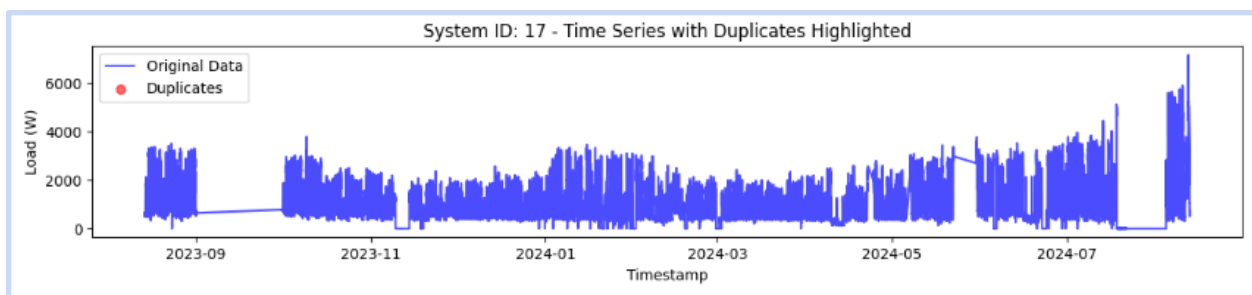
We plotted a graph to understand where the duplicates value in our data set. Electricity Loads in a **system ID** can not be the same with each changing time stamp. It may vary with changing timestamps.

EDA 1



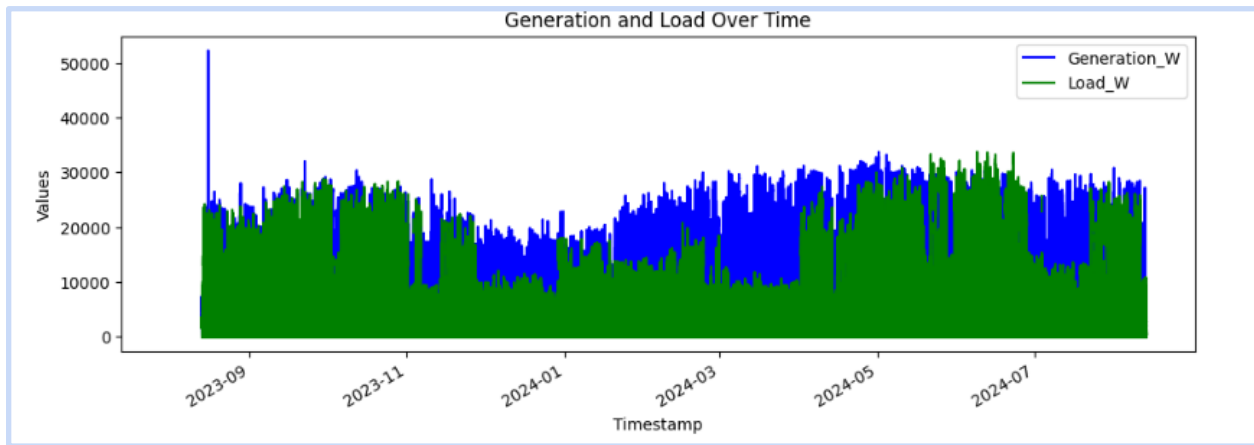
The graph shows that in lower load (W), there are a lot of duplicates which means that the sensor gauging load (W) with low load (W) was less precise.

EDA 2



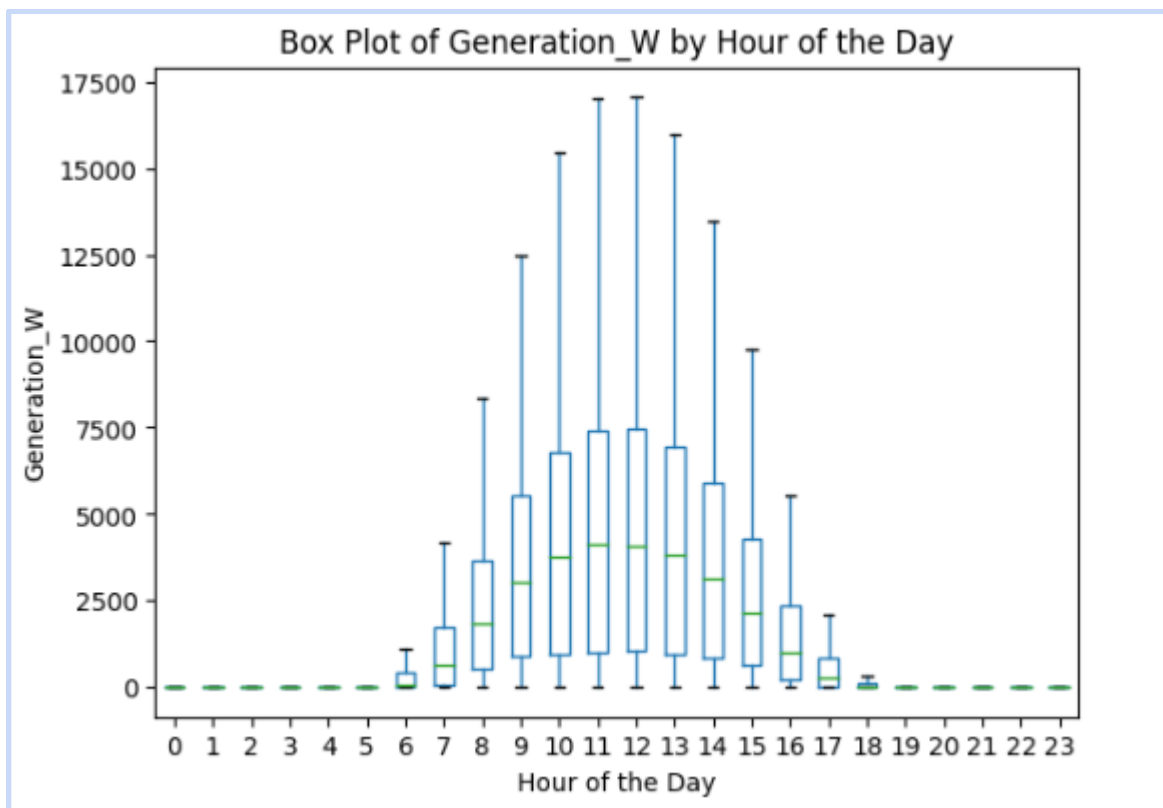
The graph shows the load(W) of **System ID 17**. We can see how the fluctuation is prominent from August-2023 till September 2024. The graphs also show that there are missing values where there is no frequency seen on the time series data.

EDA 3



There are discrepancies in the generation and load over time. Generation is high as compared to load which means there is a lot of loss of generated output which needs to be optimized.

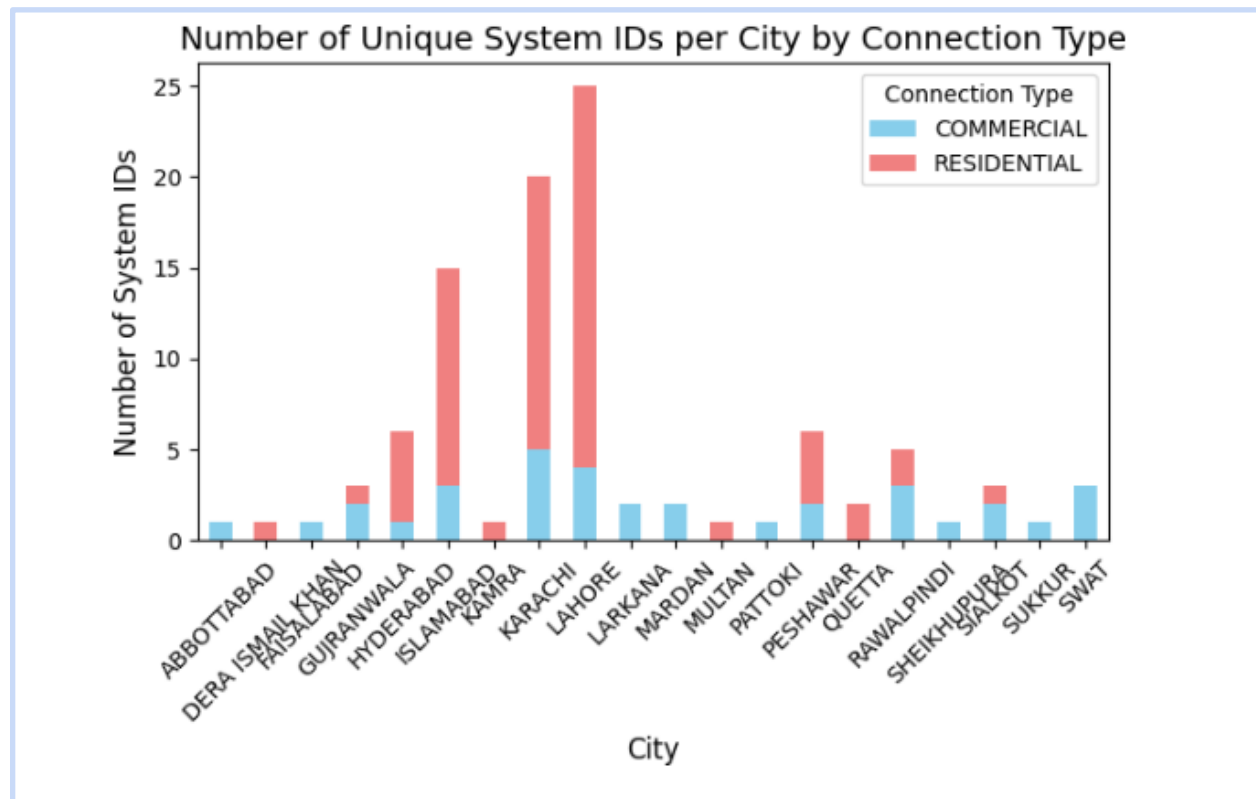
EDA 4



This graph here shows a prominent relationship between Hours of the day and the generation taking place in those hours. We can monitor that the generation starts at 6am in the morning as the sun rises and goes to maximum between 11am and 12pm as the sun emerges its energy from the top angle and later as the day starts ending the generation also starts decreasing.

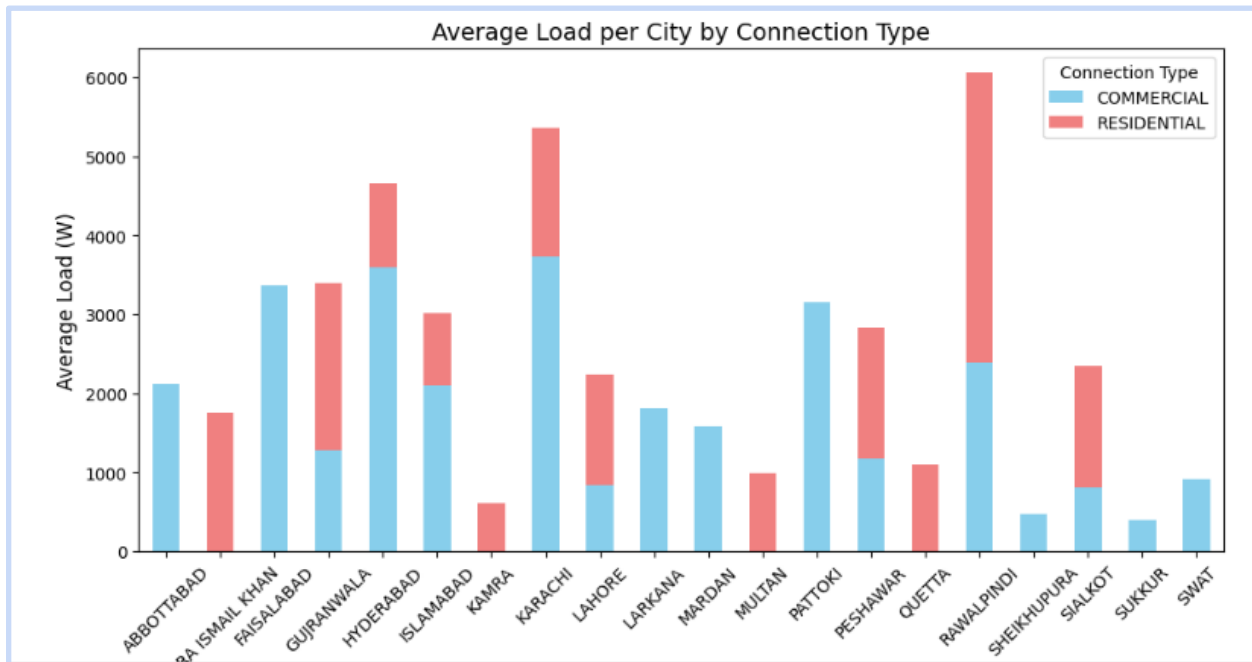
City Wise EDA

EDA 5 Number of Unique System IDs per City by Connection Type

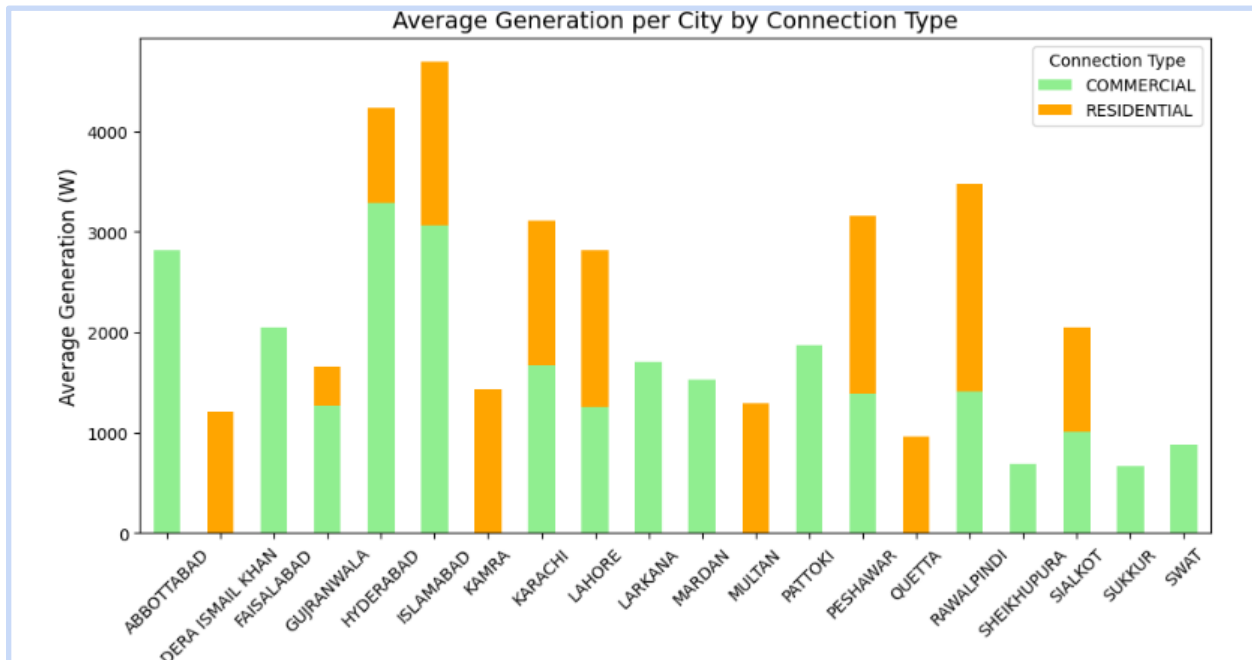


We can see that the data has most of the system IDs in Lahore, Karachi and Islamabad. Also Swat, Shukkur, Sheikhupura, Larkana, Mardan and Pattoki only have commercial connections. Quessta Mulan and Dera Ismail Khan only have residential connections.

EDA 6 Average Load per City by Connection



The graph shows the average load per city by connection type. Karachi has the most load distributed in commercial whereas Rawalpindi had most load distributed in residential sector. Sheikhupura has only commercial system IDs, but they are of small size. Even though Lahore system IDs were higher than Karachi but the



Islamabad has the highest generation of electricity. Lahore generation capacity is lower than Peshawar, Karachi and Rawalpindi. Commercial generation is higher in most of the cities except Rawalpindi, Peshawar, Lahore and Kamra.

2.2 Preprocessing of Data

Data preprocessing is an extremely important step in enhancing the performance of the model. We took the following steps to carry out preprocessing:

- 1) **Handling of Missing Values:** There were 74,298 missing values in the initial data set. To handle this issue, we used interpolation to fill in the missing values. Interpolation helped in estimating the missing values based on surrounding data, ensuring continuity in the data set to reduce any potential biases that could have seeped in due to the missing data.
- 2) **Removal of Duplicates:** Initially there were 79,551 duplicate entries in the data set that we identified and removed to prevent skewing when training the model. Duplicate records could have potentially distorted the learning process, leading to overfitting and inaccurate predictions.

First we used data manipulation techniques to check for rows that had the same values in the following fields: timestamps and system identifiers.

Then we used Pandas in python to streamline the detection of duplicates. The *drop_duplicates()* function quickly allowed us to quickly identify duplicate rows.

We set the criteria for duplication as (a) matching timestamps to ensure that records from the same timestamp were compared and (b) identical values in relevant columns to check that other significant fields such as PV generation and load consumption values were also the same.

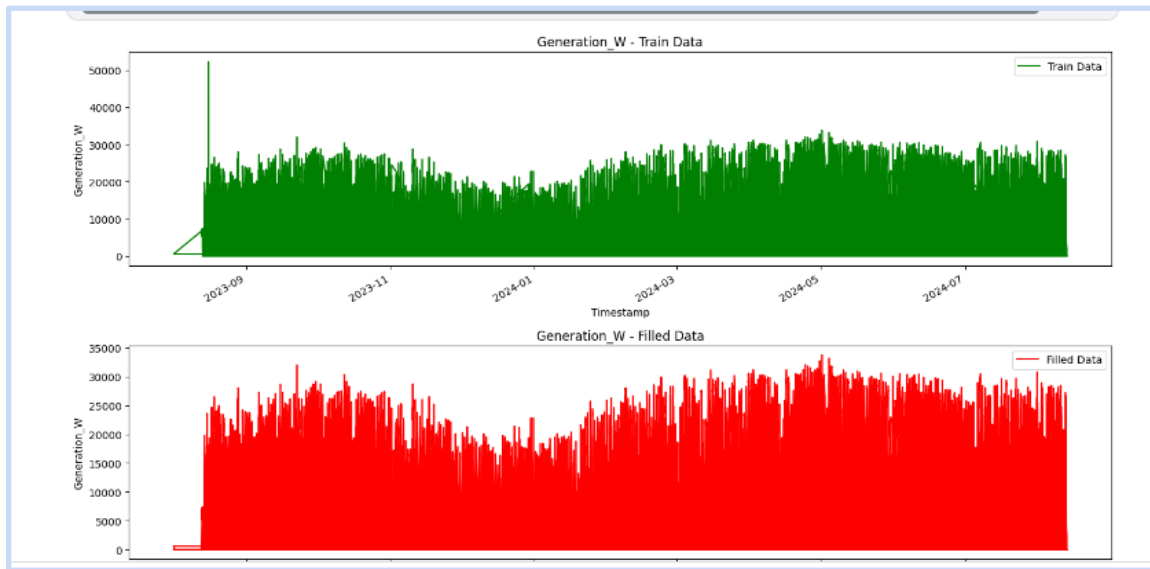
The flagged duplicates were reviewed to ensure that no valuable data was mistakenly removed and then the *drop_duplicates()* function was used to remove duplicates from the dataset.

3) Treatment of Outliers: To identify outliers, we used the Z score method and interquartile range. In addition to this, we also deployed data visualization techniques such as box plot and scatter plot to inspect the data for any anomalies. We had to make a decision whether outliers were to be removed, transformed or capped. After analysis of the data, we decided to remove outliers using Pandas in python by using boolean indexing and additionally the *clip()* function.

4) Correction of Erroneous Generation Data for Accurate Prediction: The data was incorrect, as it showed electricity generation occurring between 17:00 and 06:00 in many cases, which could negatively impact our prediction. Since no electricity is produced during these hours, we addressed the issue by assigning a generation value of zero to all data points within this time range. This adjustment aligns with the expected behavior of electricity production during these intervals.

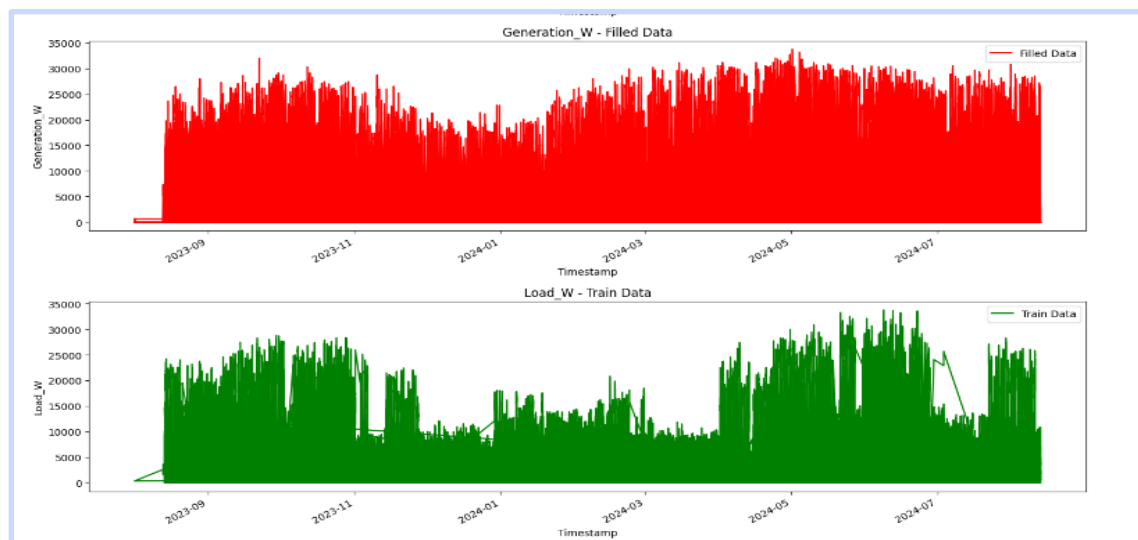
These data preprocessing steps gave us a clean data set that was ready for feature engineering and model training.

EDA 8



The graph shows the distribution of generation data before and after cleaning the data. The frequency has increased.

EDA 9



This graph shows the distribution of load data after cleaning the data and there is a significant change that can be seen in the graphs for load data. The fluctuation of load has increased.

5) One Hard Encoding: We did one hard encoding for the **Location** and **System IDs**.

2.3 Feature Engineering

This phase was taken to transform the raw timestamp data into several new features to capture the important temporal and contextual patterns. These transformations included:

1) **Temporal Features:** We converted the raw timestamp data into year, month, day, and hour of the week. These features were vital to capture the seasonal patterns that affect both PV generation and load consumption. For instance, energy generation typically peaked during midday and varied across different seasons, which we saw in different months of the year (e.g. higher generation in August which is a warm season in Pakistan generally).

2) **Weather Related Features:** We integrated features such as temperature, windspeed, and atmospheric pressure, to enhance the model's understanding of factors that could perhaps influence the level of energy generation. Higher temperatures e.g. reduce the efficiency of solar panels and varying wind speeds can impact sunlight exposure. DNI has direct relationship.

The inclusion of these features improves the models' predictive power and additionally provides essential context to understand the underlying dynamics of energy generation and consumption.

3) **Time Stamp:** We separated year, month, day, hour, and minute from the timestamp and made it features for our model. We also used sin and cos features for our model to handle the lag and to understand the frequency in a better manner.

2.4 Model Selection

We used two different machine learning models for this project: LightGBM and GRU.

1) **LightGBM:** Light Gradient Boosting is an ensemble model used to effectively work for gradient boosting. Gradient boosting means that the deep neural network model works on minimizing the cost of the model by increasing the accuracy of a big data set like in our case. LightGBM makes the accuracy high. We had to do hyper tuning of the model which helped us achieve more accuracy. We used boosting rounds and learning rates to increase the accuracy of the model.

2) **LSTM Long Short-Term Memory:** is a recurrent neural network (RNN) and is used to learn the patterns of the sequential data. It is also used to generate models. We used this approach too so that data may understand the patterns in each system IDs. The challenge using LSTM was that it had to train the data set with a high epoch level which took both high computation and a lot of time.

By selecting both of these models, we aimed to leverage the strengths of each to optimize predictions of PV generation and load consumption.

3. Results

3.1 Performance of LightGBM and LSTM

Mean Absolute error was used to reduce the loss incurred by the model. We started the MAE by 1200 and took it down to around 740. Experimentation was done to reduce the MAE. Different models and different hyper tuning techniques were also used.

We divided the data into test data and validation data. The idea behind the validation data was to not show the model data and understand with its underfit or over fit.

For LSTM we also used the same techniques, but the loss was very less because of high training and the model was over fitting.

3.2 Insights from Data

To understand the dynamics of PV generation and load consumption, we derived several key insights that were critical:

1) **Seasonality:** Features that were time based such as hour of the day and day of the week were found to be highly correlated with the PV generation patterns. We saw that the solar energy generation typically peaked during midday when the sun is at its highest and sunlight is abundant.

Consumption on the other hand fluctuates according to human activities and routines based on the hour of the day, contingent to working hours.

2) **Weather Dependencies:** Our analysis indicates that weather variables and energy generation are significantly correlated. High winds can impact solar energy production through direct and indirect means. This can be explained by the wind affecting how the clouds travel through the atmosphere. High wind speeds have the potential to cause abrupt fluctuations in cloud cover, which in turn impacts the solar panels' solar radiation levels. High wind speeds have the ability to harm the solar panels as well. Another important component affecting the production of solar energy is variations in air pressure, which has an impact on cloud formation, temperature, humidity, and long-term climatic trends, all of which are related to solar generation capacity.

These findings highlight the need for prediction models to include both environmental and temporal aspects in order to improve their relevance and accuracy.

4. Insights and Conclusion

Our model directed us towards several significant insights into the prediction of photovoltaic generation and load consumption by utilizing the LightGBM and GRU models in several different ways. An important realization we came to be the relevance of time-based features in energy generation forecasting. Our models showed that the amount of solar energy produced not only had a strong relationship with the time of day but also peaked at midday when the sun is usually at its highest. Both models proved to do a good job of capturing this cyclical pattern. This showed us how crucial it is to incorporate time-based data in prediction models such as these.

Weather-related characteristics also had a substantial impact on the performance of the models. We noted that PV generation was significantly impacted by variables such as temperature, wind speed, and air pressure. For instance, we know that lower solar panel efficiency has been linked to greater temperatures, which does eventually result in lower energy generation. On the other hand, wind speed and atmospheric conditions also affect the amount of sunlight that is received, as discussed above in the results. Hence, the PV production was affected, especially in overcast or stormy weather. When we added these weather-related factors, the capacity of both models to produce precise forecasts was greatly increased.

Although both models produced insightful results, their unique advantages opened up new areas of exploration for us. It is integral to notice that when working with huge datasets that featured complicated feature interactions, LightGBM performed significantly well. It was the perfect choice for large-scale energy forecasting applications since it had the ability to analyze structured data rapidly and quickly during training. However, our second model, GRU, was more successful in identifying the temporal connections in the data. Hence, it served as the superior choice in situations where long-term patterns were important. However, we noted that it increased computing complexity.

Overall, we concluded that both models were excellent at forecasting PV generation and load consumption. Also, depending on the particulars of the prediction assignment, each model had particular strengths.

5. Recommendation

Based on our model and insights shared above, below are a few suggestions for improving the operational effectiveness of grid stations to conduct better delivery and management of renewable energy. These are an effort towards enhancing the stability of power grids through data management and forecasting methodologies:

1) **Weather Monitoring to be Real Time:** As we aim to leverage our models more effectively, we believe that if the grid systems deploy real-time weather monitoring systems, they are likely to obtain real-time data on important factors such as solar irradiance, temperature, and wind speed. Another effective method can be to update predictive models directly with this data which can produce more precision in forecasting the energy generation and also provide grid stations with better management choices.

2) **Predictive Maintenance:** Grid stations may suffer massively due to equipment malfunction. To overcome this challenge, they may leverage machine learning models to predict when solar panels and inverters, etc. may require repair or maintenance. This is an effective and proactive strategy to reduce unexpected faults in equipment. Eventually, grid stations can achieve streamlined energy production and longer equipment life.

3) **Storage Solutions for Energy:** If grid stations start to store excess energy generated during peak solar hours, they can function more effectively. Batteries can serve this purpose and release this stored energy when there is insufficient sunlight in order to supply electricity.

4) **Introducing Dynamic Grid Systems:** Grid stations can also enhance productivity by leveraging dynamic systems for energy distribution in real time. This can be done effectively if supply and demand are balanced using predictive models. In this way, grid managers can react ahead of time to any changes in upcoming energy generation or consumption.

5) **Using Satellite and Geographic Data:** Lastly, grid stations can use data from other enhanced sources such as satellite images and geographic information to estimate solar irradiance with much more precision. This information can be useful for adjusting panels.